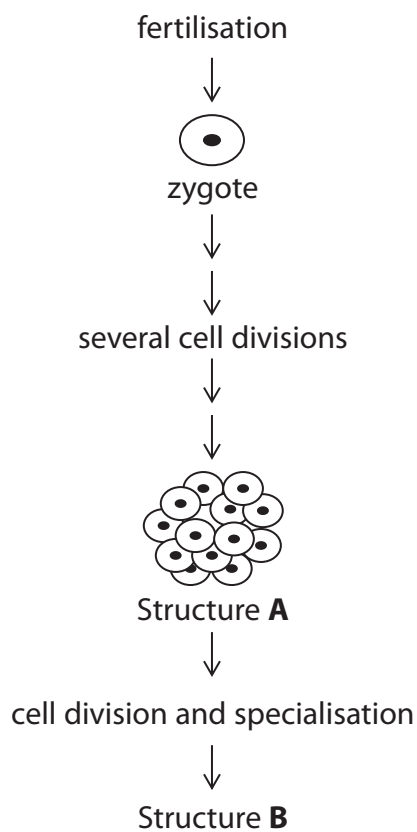


- 4 The diagram shows some processes that occur after the fertilisation of an egg cell.



- (a) Structure **B** is a blastocyst.

Describe the structure of a blastocyst.

(2)

.....

.....

.....

.....

.....

.....

- (b) Structure **A** contained 60 cells.

The mitotic index of cells from structure **A** was calculated as 15%.

Calculate the number of cells in structure **A** which were in mitosis.

(1)

Answer

- (c) Structure **B** contains pluripotent stem cells.

Histone modification and DNA methylation are processes that occur when totipotent stem cells develop into pluripotent stem cells.

Explain how these two processes alter the activation of genes in the cells.

(5)

(Total for Question 4 = 8 marks)

